

Proofs

Analysis

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IVT Proof – By Contradiction

Claim: If f is continuous over $[a, b]$, then for every $y \in [f(a), f(b)]$, there exists some x such that $f(x) = y$.

Proof: Suppose not. Then, there exists some $y \in [f(a), f(b)]$ such that for no value of $x \in [a, b]$ does $y = f(x)$.